

# Phase 1 — Data Requirements

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*StoreCraft: A Multi-Tenant E-Commerce Platform*

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## COM2058 Project — Phase 1: Data Requirements

### 1. Overview

#### 1.1 Scope of the Conceptual Design

Entity count

### 2. Tenant & Identity Model

Tenancy

USERS subtypes — modelled as IS\_A

Staff roles

### 3. Entity Attribute Dictionary

Group 1 — Identity (4 entities)

Group 2 — Tenant + Catalog (4 entities)

Group 3 — Inventory (1 entity + 1 ternary relationship)

Group 4 — Commerce (5 entities)

Group 5 — Engagement + Audit (3 entities)

### 4. Relationships (Summary)

### 5. Business Rules & Constraints

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## 1. Overview

StoreCraft is a multi-tenant SaaS platform that lets independent merchants launch online stores. Each merchant (tenant) maintains its own catalog, customer base, orders, and inventory; data is isolated at the schema level. End-users are modeled globally via three binary **IS\_A** (1:1) subtype relationships so one person can shop across many merchant stores with a single identity.

### 1.1 Scope of the Conceptual Design

The conceptual data model presented in this document and refined in the Phase 2 ER diagram covers the full operational lifecycle of a multi-tenant e-commerce platform. The functional domains in scope are formalised as follows:

1. **Catalog management.** Products, product variants (modelled as a weak entity dependent on its parent product), and a recursive category hierarchy under each merchant.
2. **Warehouse and inventory control.** Per-merchant warehouses and stock levels modelled as a genuine ternary relationship **STOCKED\_AT(PRODUCTS × PRODUCT\_VARIANTS × WAREHOUSES)**, with derived availability quantities.
3. **Shopping workflow.** Customer shopping carts (including guest carts), finalised orders, and order line items (modelled as a second weak entity dependent on its order).
4. **Order fulfilment.** Payments and shipments associated with each order, with one-to-many cardinality on both legs to permit partial captures and split fulfilment.
5. **Customer engagement.** Product reviews authored by customers and merchant-issued discount codes redeemable on orders.
6. **Auditability.** An append-only activity log implementing a polymorphic association over the auditable entity set.

### Entity count

**17 entity types** in the ER diagram (4 Identity + 4 Tenant/Catalog + 1 Warehouses + 5 Commerce + 3 Engagement). **INVENTORY** -> **ternary relationship** **STOCKED\_AT(PRODUCTS × PRODUCT\_VARIANTS × WAREHOUSES)** with attributes, not a standalone entity. During Phase 3 relational mapping, the ternary becomes its own bridge table — yielding 18 total tables. Additional bridge tables (**merchant\_staff**, **product\_categories**, **cart\_items**, **discount\_usages**) also emerge then.

## 2. Tenant & Identity Model

### Tenancy

- **Tenant root:** `MERCHANTS` — each row = one store owner / shop
- **Tenant-scoped:** every catalog/commerce/inventory/audit table carries `merchant_id`
- **Tenant-free (global):** `USERS` and its three subtype entities (`CUSTOMER`, `STAFF`, `PLATFORM_ADMIN`)

### USERS subtypes — modelled as IS\_A

- **Subtypes:** `CUSTOMER`, `STAFF`, `PLATFORM_ADMIN` (3 separate entities)
- **Subtype rows may coexist:** one `USERS` row may simultaneously have a `CUSTOMER` row *and* a `STAFF` row (e.g., a merchant owner who also shops on other stores).
- **Subtype rows are optional:** a freshly-registered user may have no subtype row at all.
- **Notation:** — three binary 1:1 `IS_A` relationships. The EER `od/oo` specialization circle from Ch08 is **not** used.

```
USERS —(0,1)— IS_A_CUST —(1,1)— CUSTOMER
USERS —(0,1)— IS_A_STAFF —(1,1)— STAFF
USERS —(0,1)— IS_A_PA —(1,1)— PLATFORM_ADMIN
```

### Staff roles

Stored as a plain string on the `merchant_staff` bridge (4 values): `owner`, `admin`, `staff`, `viewer`. RBAC/permission tables deferred to Phase 4+.

## 3. Entity Attribute Dictionary

### Group 1 — Identity (4 entities)

#### 3.1 USERS (base entity)

Shared attributes every authenticated person carries. The three `IS_A` subtypes (`CUSTOMER`, `STAFF`, `PLATFORM_ADMIN`) reference this row.

| Attribute                      | Type         | Key / Null             | Notes                     |
|--------------------------------|--------------|------------------------|---------------------------|
| <code>user_id</code>           | BIGINT       | PK                     | Surrogate, auto-increment |
| <code>email</code>             | VARCHAR(255) | UNIQUE, NOT NULL       | Login credential          |
| <code>password_hash</code>     | VARCHAR(255) | NOT NULL               | Argon2id or bcrypt        |
| <code>first_name</code>        | VARCHAR(80)  | NOT NULL               |                           |
| <code>last_name</code>         | VARCHAR(80)  | NOT NULL               |                           |
| <code>phone</code>             | VARCHAR(20)  | NULL                   | E.164 format              |
| <code>email_verified_at</code> | DATETIME     | NULL                   | Non-null = verified       |
| <code>is_active</code>         | BOOLEAN      | NOT NULL, default TRUE | Soft-disable flag         |
| <code>last_login_at</code>     | DATETIME     | NULL                   |                           |
| <code>created_at</code>        | DATETIME     | NOT NULL               |                           |
| <code>updated_at</code>        | DATETIME     | NOT NULL               |                           |

### 3.2 CUSTOMER (IS\_A subtype of USERS)

Subtype of USERS via the IS\_A\_CUST relationship — users who shop on any merchant's store.

| Attribute                             | Type        | Key / Null              | Notes                                                                                                              |
|---------------------------------------|-------------|-------------------------|--------------------------------------------------------------------------------------------------------------------|
| <code>user_id</code>                  | BIGINT      | PK, FK → USERS          | Shared PK — IS_A subtype reference                                                                                 |
| <code>default_shipping_address</code> | COMPOSITE   | NULL                    | { <code>street</code> , <code>city</code> , <code>state</code> , <code>postal_code</code> , <code>country</code> } |
| <code>date_of_birth</code>            | DATE        | NULL                    |                                                                                                                    |
| <code>loyalty_points</code>           | INT         | NOT NULL, default 0     | Across all merchants (global balance)                                                                              |
| <code>accepts_marketing</code>        | BOOLEAN     | NOT NULL, default FALSE | Global opt-in                                                                                                      |
| <code>referral_code</code>            | VARCHAR(20) | UNIQUE, NULL            |                                                                                                                    |

### 3.3 STAFF (IS\_A subtype of USERS)

Subtype of USERS via the IS\_A\_STAFF relationship — users employed at one or more merchants.

| Attribute                      | Type         | Key / Null     | Notes                                                                     |
|--------------------------------|--------------|----------------|---------------------------------------------------------------------------|
| <code>user_id</code>           | BIGINT       | PK, FK → USERS | Shared PK — IS_A subtype reference                                        |
| <code>employment_type</code>   | VARCHAR(20)  | NOT NULL       | <code>full_time</code> / <code>part_time</code> / <code>contractor</code> |
| <code>hired_at</code>          | DATETIME     | NOT NULL       | First-employment date across StoreCraft                                   |
| <code>title</code>             | VARCHAR(80)  | NOT NULL       | e.g. “Store Manager”, “Sales Assistant”                                   |
| <code>commission_rate</code>   | DECIMAL(5,2) | NULL           | % of sales; null if salaried                                              |
| <code>employment_status</code> | VARCHAR(20)  | NOT NULL       | <code>active</code> / <code>on_leave</code> / <code>terminated</code>     |

### 3.4 PLATFORM\_ADMIN (IS\_A subtype of USERS)

Subtype of USERS via the IS\_A\_PA relationship — StoreCraft platform employees (our own team).

| Attribute                | Type        | Key / Null     | Notes                                                                                          |
|--------------------------|-------------|----------------|------------------------------------------------------------------------------------------------|
| <code>user_id</code>     | BIGINT      | PK, FK → USERS | Shared PK — IS_A subtype reference                                                             |
| <code>admin_level</code> | VARCHAR(20) | NOT NULL       | <code>super_admin</code> / <code>support</code> / <code>engineer</code> / <code>billing</code> |
| <code>department</code>  | VARCHAR(50) | NOT NULL       | e.g. “Customer Success”, “Trust & Safety”                                                      |
| <code>hired_at</code>    | DATETIME    | NOT NULL       |                                                                                                |

## Group 2 — Tenant + Catalog (4 entities)

### 3.5 MERCHANTS (tenant root)

One row = one store = one tenant. All catalog/commerce data below carries `merchant_id`.

| Attribute                     | Type         | Key / Null           | Notes                                                                |
|-------------------------------|--------------|----------------------|----------------------------------------------------------------------|
| <code>merchant_id</code>      | BIGINT       | PK                   | Tenant identifier                                                    |
| <code>slug</code>             | VARCHAR(64)  | UNIQUE, NOT NULL     | URL slug<br>( <code>storecraft.com/{slug}</code> )                   |
| <code>store_name</code>       | VARCHAR(120) | NOT NULL             | Public display name                                                  |
| <code>owner_user_id</code>    | BIGINT       | FK → STAFF, NOT NULL | Founding staff (owner)                                               |
| <code>business_address</code> | COMPOSITE    | NOT NULL             | <code>{street, city, state, postal_code, country}</code> —           |
| <code>contact_email</code>    | VARCHAR(255) | NOT NULL             | Support email                                                        |
| <code>currency</code>         | CHAR(3)      | NOT NULL             | ISO 4217 (USD, TRY, EUR)                                             |
| <code>plan</code>             | VARCHAR(20)  | NOT NULL             | <code>starter</code> / <code>growth</code> / <code>enterprise</code> |
| <code>created_at</code>       | DATETIME     | NOT NULL             |                                                                      |
| <code>activated_at</code>     | DATETIME     | NULL                 | When plan payment confirmed                                          |
| <code>suspended_at</code>     | DATETIME     | NULL                 | Non-null = suspended                                                 |

### 3.6 PRODUCTS

| Attribute                 | Type          | Key / Null               | Notes                                                                                                                                   |
|---------------------------|---------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <code>product_id</code>   | BIGINT        | PK                       |                                                                                                                                         |
| <code>merchant_id</code>  | BIGINT        | FK → MERCHANTS, NOT NULL | Tenant anchor                                                                                                                           |
| <code>slug</code>         | VARCHAR(120)  | NOT NULL                 | UNIQUE ( <code>merchant_id</code> , <code>slug</code> )                                                                                 |
| <code>title</code>        | VARCHAR(255)  | NOT NULL                 |                                                                                                                                         |
| <code>description</code>  | TEXT          | NULL                     |                                                                                                                                         |
| <code>product_type</code> | VARCHAR(20)   | NOT NULL                 | Enum kept as plain attribute: <code>physical</code> / <code>digital</code> / <code>subscription</code> (no subtype entities in Phase 2) |
| <code>base_price</code>   | DECIMAL(12,2) | NOT NULL                 | Variants may override                                                                                                                   |
| <code>currency</code>     | CHAR(3)       | NOT NULL                 |                                                                                                                                         |
| <code>status</code>       | VARCHAR(20)   | NOT NULL                 | <code>draft</code> / <code>active</code> / <code>archived</code>                                                                        |
| <code>created_at</code>   | DATETIME      | NOT NULL                 |                                                                                                                                         |
| <code>updated_at</code>   | DATETIME      | NOT NULL                 |                                                                                                                                         |

### 3.7 PRODUCT\_VARIANTS (weak entity)

No standalone identity — a variant only exists under a product.



| Attribute                   | Type          | Key / Null                                                          | Notes                                    |
|-----------------------------|---------------|---------------------------------------------------------------------|------------------------------------------|
| <code>product_id</code>     | BIGINT        | PK (part), FK → PRODUCTS                                            | Identifying relationship                 |
| <code>variant_no</code>     | INT           | PK (part)                                                           | Partial key — sequenced within a product |
| <code>sku</code>            | VARCHAR(40)   | UNIQUE<br>( <code>merchant_id</code> , <code>sku</code> ), NOT NULL | Stock-keeping unit                       |
| <code>option1_name</code>   | VARCHAR(40)   | NULL                                                                | e.g. “Color”                             |
| <code>option1_value</code>  | VARCHAR(40)   | NULL                                                                | e.g. “Red”                               |
| <code>option2_name</code>   | VARCHAR(40)   | NULL                                                                | e.g. “Size”                              |
| <code>option2_value</code>  | VARCHAR(40)   | NULL                                                                | e.g. “L”                                 |
| <code>price_override</code> | DECIMAL(12,2) | NULL                                                                | NULL = use product.base_price            |
| <code>barcode</code>        | VARCHAR(64)   | NULL                                                                | EAN / UPC                                |
| <code>is_default</code>     | BOOLEAN       | NOT NULL, default FALSE                                             | Default variant flag                     |

Weak entity semantics: Compound PK (`product_id`, `variant_no`) ;

### 3.8 CATEGORIES (recursive hierarchy)

| Attribute                       | Type         | Key / Null               | Notes                                                      |
|---------------------------------|--------------|--------------------------|------------------------------------------------------------|
| <code>category_id</code>        | BIGINT       | PK                       |                                                            |
| <code>merchant_id</code>        | BIGINT       | FK → MERCHANTS, NOT NULL | Tenant anchor                                              |
| <code>parent_category_id</code> | BIGINT       | FK → CATEGORIES, NULL    | Recursive self-FK                                          |
| <code>slug</code>               | VARCHAR(80)  | NOT NULL                 | UNIQUE<br>( <code>merchant_id</code> , <code>slug</code> ) |
| <code>name</code>               | VARCHAR(120) | NOT NULL                 |                                                            |
| <code>display_order</code>      | SMALLINT     | NOT NULL, default 0      | Sibling ordering                                           |
| <code>created_at</code>         | DATETIME     | NOT NULL                 |                                                            |

**Recursive relationship:** `parent_category_id` → `category_id` creates a tree (e.g., Electronics → Phones → Smartphones).

## Group 3 — Inventory (1 entity + 1 ternary relationship)

### 3.9 WAREHOUSES

Physical storage locations. Each warehouse belongs to one merchant.

| Attribute                 | Type        | Key / Null               | Notes                                                                                                              |
|---------------------------|-------------|--------------------------|--------------------------------------------------------------------------------------------------------------------|
| <code>warehouse_id</code> | BIGINT      | PK                       |                                                                                                                    |
| <code>merchant_id</code>  | BIGINT      | FK → MERCHANTS, NOT NULL | Tenant anchor                                                                                                      |
| <code>name</code>         | VARCHAR(80) | NOT NULL                 | e.g., “Ankara Main Warehouse”                                                                                      |
| <code>address</code>      | COMPOSITE   | NOT NULL                 | { <code>street</code> , <code>city</code> , <code>state</code> , <code>postal_code</code> , <code>country</code> } |
| <code>is_active</code>    | BOOLEAN     | NOT NULL, default TRUE   | Operational flag                                                                                                   |
| <code>created_at</code>   | DATETIME    | NOT NULL                 |                                                                                                                    |

### 3.10 INVENTORY (ternary relationship: PRODUCT × VARIANT × WAREHOUSE)

Genuine ternary — quantity is defined only by the triple of (product, variant, warehouse). Binary decomposition loses information.

| Attribute                       | Type     | Key / Null                                   | Notes                                          |
|---------------------------------|----------|----------------------------------------------|------------------------------------------------|
| <code>product_id</code>         | BIGINT   | PK (part), FK → PRODUCTS                     | Ternary participant 1                          |
| <code>variant_no</code>         | INT      | PK (part), FK → PRODUCT_VARIANTS (composite) | Ternary participant 2                          |
| <code>warehouse_id</code>       | BIGINT   | PK (part), FK → WAREHOUSES                   | Ternary participant 3                          |
| <code>merchant_id</code>        | BIGINT   | FK → MERCHANTS, NOT NULL                     | Tenant anchor (composite-FK enforcement)       |
| <code>quantity_on_hand</code>   | INT      | NOT NULL, default 0                          | Physical stock                                 |
| <code>quantity_reserved</code>  | INT      | NOT NULL, default 0                          | Reserved by carts/orders                       |
| <code>quantity_available</code> | INT      | DERIVED                                      | = <code>on_hand</code> – <code>reserved</code> |
| <code>reorder_level</code>      | INT      | NULL                                         | Low-stock threshold                            |
| <code>last_restocked_at</code>  | DATETIME | NULL                                         |                                                |
| <code>updated_at</code>         | DATETIME | NOT NULL                                     |                                                |

**Compound PK:** (`product_id`, `variant_no`, `warehouse_id`).



**Why genuine ternary (vs. binary decomposition):** a binary `variant↔warehouse` paired with `variant↔product` cannot express “variant X in warehouse Y has N units” — the count depends on all three. This argument is expanded in the Phase 4 report.

## Group 4 — Commerce (5 entities)

### 3.11 CARTS

Active shopping carts. A cart lives under one merchant’s storefront.

| Attribute                     | Type        | Key / Null               | Notes                                                                 |
|-------------------------------|-------------|--------------------------|-----------------------------------------------------------------------|
| <code>cart_id</code>          | BIGINT      | PK                       |                                                                       |
| <code>merchant_id</code>      | BIGINT      | FK → MERCHANTS, NOT NULL | Tenant anchor                                                         |
| <code>customer_user_id</code> | BIGINT      | FK → CUSTOMER, NULL      | NULL for guest carts                                                  |
| <code>session_token</code>    | VARCHAR(64) | NULL                     | Guest-cart identifier                                                 |
| <code>currency</code>         | CHAR(3)     | NOT NULL                 | Frozen at cart creation                                               |
| <code>status</code>           | VARCHAR(20) | NOT NULL                 | <code>active</code> / <code>abandoned</code> / <code>converted</code> |
| <code>expires_at</code>       | DATETIME    | NULL                     | GC for abandoned carts                                                |
| <code>created_at</code>       | DATETIME    | NOT NULL                 |                                                                       |
| <code>updated_at</code>       | DATETIME    | NOT NULL                 |                                                                       |

Cart-line data will surface in Phase 2 as the `cart_items` bridge (M:N: carts  $\square$  product\_variants, with `quantity` attribute).

### 3.12 ORDERS

Finalized purchases. Immutable once placed (revisions are handled via refund/cancel).

| Attribute                     | Type             | Key / Null                                                                | Notes                                                                                                                                    |
|-------------------------------|------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <code>order_id</code>         | BIGINT           | <b>PK</b>                                                                 |                                                                                                                                          |
| <code>merchant_id</code>      | BIGINT           | FK → MERCHANTS, NOT NULL                                                  | Tenant anchor                                                                                                                            |
| <code>customer_user_id</code> | BIGINT           | FK → CUSTOMER, NOT NULL                                                   | Registered customer required                                                                                                             |
| <code>order_number</code>     | VARCHAR(20)      | UNIQUE ( <code>merchant_id</code> , <code>order_number</code> ), NOT NULL | Public ID ( <code>SC-10001</code> )                                                                                                      |
| <code>status</code>           | VARCHAR(20)      | NOT NULL                                                                  | <code>pending</code> / <code>paid</code> / <code>shipped</code> / <code>delivered</code> / <code>canceled</code> / <code>refunded</code> |
| <code>shipping_address</code> | <b>COMPOSITE</b> | NOT NULL                                                                  | Snapshot of customer address                                                                                                             |
| <code>billing_address</code>  | <b>COMPOSITE</b> | NOT NULL                                                                  | May differ from shipping                                                                                                                 |
| <code>subtotal</code>         | DECIMAL(12,2)    | NOT NULL                                                                  | Sum of line items                                                                                                                        |
| <code>discount_total</code>   | DECIMAL(12,2)    | NOT NULL, default 0                                                       | Applied coupons                                                                                                                          |
| <code>tax_total</code>        | DECIMAL(12,2)    | NOT NULL, default 0                                                       |                                                                                                                                          |
| <code>shipping_total</code>   | DECIMAL(12,2)    | NOT NULL, default 0                                                       | Carrier fees                                                                                                                             |
| <code>grand_total</code>      | DECIMAL(12,2)    | <b>DERIVED</b>                                                            | <code>subtotal - discount_total + tax_total + shipping_total</code>                                                                      |
| <code>currency</code>         | CHAR(3)          | NOT NULL                                                                  |                                                                                                                                          |
| <code>placed_at</code>        | DATETIME         | NOT NULL                                                                  |                                                                                                                                          |
| <code>canceled_at</code>      | DATETIME         | NULL                                                                      |                                                                                                                                          |

**Address snapshots:** customer addresses can change over time; orders freeze them for accounting immutability.

### 3.13 ORDER\_ITEMS (weak entity — #2)

Line items exist only under their order.

| Attribute                    | Type          | Key / Null                                  | Notes                                 |
|------------------------------|---------------|---------------------------------------------|---------------------------------------|
| <code>order_id</code>        | BIGINT        | PK (part), FK → ORDERS                      | Identifying relationship              |
| <code>line_no</code>         | INT           | PK (part)                                   | Partial key — 1, 2, 3... within order |
| <code>product_id</code>      | BIGINT        | FK → PRODUCTS, NOT NULL                     |                                       |
| <code>variant_no</code>      | INT           | FK → PRODUCT_VARIANTS (composite), NOT NULL |                                       |
| <code>merchant_id</code>     | BIGINT        | FK → MERCHANTS, NOT NULL                    | Tenant anchor                         |
| <code>product_title</code>   | VARCHAR(255)  | NOT NULL                                    | Snapshot — frozen at order time       |
| <code>variant_label</code>   | VARCHAR(120)  | NOT NULL                                    | Snapshot — e.g., “Red / L”            |
| <code>sku</code>             | VARCHAR(40)   | NOT NULL                                    | Snapshot                              |
| <code>unit_price</code>      | DECIMAL(12,2) | NOT NULL                                    | Snapshot                              |
| <code>quantity</code>        | INT           | NOT NULL                                    |                                       |
| <code>line_subtotal</code>   | DECIMAL(12,2) | DERIVED                                     | <code>unit_price × quantity</code>    |
| <code>discount_amount</code> | DECIMAL(12,2) | NOT NULL, default 0                         | Line-level discount                   |

**Snapshot pattern:** `product_title`, `variant_label`, `sku`, `unit_price` freeze values — if the catalog later changes, order history stays intact. Phase 4 report discusses this controlled denormalization.

### 3.14 PAYMENTS

One order may have several payments (partial, retries, post-refund re-charge) → 1:N.

| Attribute         | Type          | Key / Null               | Notes                                                                            |
|-------------------|---------------|--------------------------|----------------------------------------------------------------------------------|
| payment_id        | BIGINT        | PK                       |                                                                                  |
| order_id          | BIGINT        | FK → ORDERS, NOT NULL    |                                                                                  |
| merchant_id       | BIGINT        | FK → MERCHANTS, NOT NULL | Tenant anchor                                                                    |
| payment_method    | VARCHAR(20)   | NOT NULL                 | credit_card /<br>debit_card /<br>bank_transfer /<br>cash_on_delivery /<br>wallet |
| amount            | DECIMAL(12,2) | NOT NULL                 |                                                                                  |
| currency          | CHAR(3)       | NOT NULL                 |                                                                                  |
| status            | VARCHAR(20)   | NOT NULL                 | pending /<br>authorized /<br>captured / failed /<br>refunded                     |
| gateway_reference | VARCHAR(120)  | NULL                     | Stripe / iyzico transaction ID                                                   |
| processed_at      | DATETIME      | NULL                     | Gateway confirmation time                                                        |
| created_at        | DATETIME      | NOT NULL                 |                                                                                  |

### 3.15 SHIPMENTS

One order may be split across warehouses → 1:N.

| Attribute        | Type        | Key / Null                | Notes                                                               |
|------------------|-------------|---------------------------|---------------------------------------------------------------------|
| shipment_id      | BIGINT      | PK                        |                                                                     |
| order_id         | BIGINT      | FK → ORDERS, NOT NULL     |                                                                     |
| merchant_id      | BIGINT      | FK → MERCHANTS, NOT NULL  | Tenant anchor                                                       |
| warehouse_id     | BIGINT      | FK → WAREHOUSES, NOT NULL | Origin warehouse                                                    |
| carrier          | VARCHAR(40) | NOT NULL                  | “PTT”, “Yurtiçi”, “Aras”, “DHL”, ...                                |
| tracking_number  | VARCHAR(80) | NULL                      |                                                                     |
| status           | VARCHAR(20) | NOT NULL                  | preparing /<br>shipped /<br>in_transit /<br>delivered /<br>returned |
| shipping_address | COMPOSITE   | NOT NULL                  | Snapshot                                                            |
| shipped_at       | DATETIME    | NULL                      |                                                                     |
| delivered_at     | DATETIME    | NULL                      |                                                                     |
| created_at       | DATETIME    | NOT NULL                  |                                                                     |

## Group 5 — Engagement + Audit (3 entities)

### 3.16 REVIEWS

| Attribute                         | Type         | Key / Null               | Notes                                                                 |
|-----------------------------------|--------------|--------------------------|-----------------------------------------------------------------------|
| <code>review_id</code>            | BIGINT       | PK                       |                                                                       |
| <code>product_id</code>           | BIGINT       | FK → PRODUCTS, NOT NULL  |                                                                       |
| <code>customer_user_id</code>     | BIGINT       | FK → CUSTOMER, NOT NULL  |                                                                       |
| <code>merchant_id</code>          | BIGINT       | FK → MERCHANTS, NOT NULL | Tenant anchor                                                         |
| <code>order_id</code>             | BIGINT       | FK → ORDERS, NULL        | Non-null = verified purchase                                          |
| <code>rating</code>               | SMALLINT     | NOT NULL                 | CHECK BETWEEN 1 AND 5                                                 |
| <code>title</code>                | VARCHAR(120) | NULL                     | Optional headline                                                     |
| <code>body</code>                 | TEXT         | NULL                     | Review text                                                           |
| <code>is_verified_purchase</code> | BOOLEAN      | NOT NULL, default FALSE  | TRUE iff <code>order_id</code> IS NOT NULL                            |
| <code>helpful_count</code>        | INT          | NOT NULL, default 0      | Community vote tally                                                  |
| <code>status</code>               | VARCHAR(20)  | NOT NULL                 | <code>pending</code> / <code>published</code> / <code>rejected</code> |
| <code>created_at</code>           | DATETIME     | NOT NULL                 |                                                                       |
| <code>moderated_at</code>         | DATETIME     | NULL                     |                                                                       |
| <code>moderated_by</code>         | BIGINT       | FK → STAFF, NULL         | Reviewer (self-reference to staff)                                    |

**Business rule:** One customer can review a product at most once → `UNIQUE (product_id, customer_user_id)`.



## 3.17 DISCOUNTS

| Attribute                          | Type          | Key / Null                                                              | Notes                                                                                  |
|------------------------------------|---------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <code>discount_id</code>           | BIGINT        | PK                                                                      |                                                                                        |
| <code>merchant_id</code>           | BIGINT        | FK → MERCHANTS, NOT NULL                                                | Tenant anchor                                                                          |
| <code>code</code>                  | VARCHAR(40)   | UNIQUE<br>( <code>merchant_id</code> ,<br><code>code</code> ), NOT NULL | e.g., “SUMMER20”                                                                       |
| <code>discount_type</code>         | VARCHAR(20)   | NOT NULL                                                                | <code>percentage</code> /<br><code>fixed_amount</code> /<br><code>free_shipping</code> |
| <code>value</code>                 | DECIMAL(12,2) | NOT NULL                                                                | <code>20</code> → 20% or 20 TL<br>(depending on type)                                  |
| <code>min_order_amount</code>      | DECIMAL(12,2) | NULL                                                                    | Minimum cart subtotal                                                                  |
| <code>max_uses</code>              | INT           | NULL                                                                    | NULL = unlimited                                                                       |
| <code>max_uses_per_customer</code> | INT           | NULL, default 1                                                         | Per-customer cap                                                                       |
| <code>used_count</code>            | INT           | NOT NULL, default 0                                                     | Increments as redeemed                                                                 |
| <code>starts_at</code>             | DATETIME      | NOT NULL                                                                |                                                                                        |
| <code>ends_at</code>               | DATETIME      | NULL                                                                    | NULL = open-ended                                                                      |
| <code>is_active</code>             | BOOLEAN       | NOT NULL, default TRUE                                                  | Manual kill-switch                                                                     |
| <code>created_at</code>            | DATETIME      | NOT NULL                                                                |                                                                                        |
| <code>created_by</code>            | BIGINT        | FK → STAFF, NOT NULL                                                    | Author                                                                                 |

Phase 2 will surface the `discount_usages(discount_id, order_id, used_at)` bridge (M:N: which orders used which coupon).

### 3.18 ACTIVITY\_LOG (polymorphic audit)

| Attribute     | Type         | Key / Null               | Notes                                                          |
|---------------|--------------|--------------------------|----------------------------------------------------------------|
| event_id      | BIGINT       | PK                       |                                                                |
| merchant_id   | BIGINT       | FK → MERCHANTS, NOT NULL | Tenant anchor                                                  |
| actor_user_id | BIGINT       | FK → USERS, NULL         | NULL for system events                                         |
| actor_type    | VARCHAR(20)  | NOT NULL                 | staff / customer / platform_admin / system                     |
| entity_type   | VARCHAR(40)  | NOT NULL                 | Polymorphic discriminator                                      |
| entity_id     | BIGINT       | NOT NULL                 | No FK — entity_id targets vary by entity_type                  |
| action        | VARCHAR(40)  | NOT NULL                 | created / updated / deleted / status_changed / logged_in / ... |
| payload_json  | JSON         | NULL                     | Before/after diff or action details                            |
| ip_address    | VARCHAR(45)  | NULL                     | IPv4 or IPv6                                                   |
| user_agent    | VARCHAR(255) | NULL                     | Client info                                                    |
| occurred_at   | DATETIME     | NOT NULL                 |                                                                |

**Polymorphic association:** (entity\_type, entity\_id) references different tables at runtime; no DB-level FK. This is a **controlled denormalization** — single uniform audit table at the cost of referential integrity. Phase 4 report discusses the trade-off.

## 4. Relationships (Summary)

The ER diagram (Phase 2) draws **26 relationships** (R1-R25 + R23b — REVIEWS participates in two binary relationships, one with PRODUCTS and one with CUSTOMER). Quick inventory:

| #   | Relationship                                                       | Cardinality                       | Participation (L / R)                         | Notes                                                     |
|-----|--------------------------------------------------------------------|-----------------------------------|-----------------------------------------------|-----------------------------------------------------------|
| R1  | USERS — IS_A — CUSTOMER                                            | 1:1 IS_A                          | (0,1) / (1,1)                                 | Ch07 ER binary; R1/R2/R3 may coexist for one USERS row    |
| R2  | USERS — IS_A — STAFF                                               | 1:1 IS_A                          | (0,1) / (1,1)                                 | Ch07 ER binary                                            |
| R3  | USERS — IS_A — PLATFORM_ADMIN                                      | 1:1 IS_A                          | (0,1) / (1,1)                                 | Ch07 ER binary                                            |
| R4  | STAFF — WORKS_FOR — MERCHANTS (via merchant_staff)                 | M:N                               | (0,N) partial / (1,N) <b>total</b> (≥1 owner) | Carries <b>role</b> attribute                             |
| R5  | STAFF — OWNER_OF — MERCHANTS                                       | 1:1 (via merchants.owner_user_id) | (0,1) partial / (1,1) <b>total</b>            | Founding owner — every merchant must have one staff owner |
| R6  | MERCHANTS — HAS — PRODUCTS                                         | 1:N                               | (0,N) partial / (1,1) <b>total</b>            |                                                           |
| R7  | MERCHANTS — HAS — CATEGORIES                                       | 1:N                               | (0,N) partial / (1,1) <b>total</b>            |                                                           |
| R8  | MERCHANTS — HAS — WAREHOUSES                                       | 1:N                               | (0,N) partial / (1,1) <b>total</b>            |                                                           |
| R9  | CATEGORIES — PARENT_OF — CATEGORIES                                | 1:N recursive                     | (0,N) partial / (0,1) partial                 | Tree                                                      |
| R10 | PRODUCTS — CATEGORIZED_IN — CATEGORIES (bridge product_categories) | M:N                               | (0,N) partial / (0,N) partial                 |                                                           |
| R11 | PRODUCTS — HAS_VARIANT — PRODUCT_VARIANTS (weak/identifying)       | 1:N identifying                   | (0,N) partial / (1,1) <b>total</b>            |                                                           |
| R12 | STOCKED_AT (PRODUCTS × PRODUCT_VARIANTS × WAREHOUSES)              | <b>Ternary</b> M:N:N              | all (0,N) partial                             | Genuine ternary                                           |
| R13 | MERCHANTS — HAS — CARTS                                            | 1:N                               | (0,N) partial / (1,1) <b>total</b>            |                                                           |
| R14 | CUSTOMER — SHOPS_AT — CARTS                                        | 1:N                               |                                               |                                                           |

| #    | Relationship                                                                            | Cardinality     | Participation (L / R)                    | Notes                                                                                |
|------|-----------------------------------------------------------------------------------------|-----------------|------------------------------------------|--------------------------------------------------------------------------------------|
|      |                                                                                         |                 | (0,N)<br>partial /<br>(0,1) partial      | CART side (0,1): guest<br>cart has null<br><b>customer_user_id</b>                   |
| R15  | <b>CARTS – CART_ITEMS</b><br>– <b>PRODUCT_VARIANTS</b><br>(bridge <b>cart_items</b> )   | M:N             | (0,N)<br>partial /<br>(0,N) partial      | <b>quantity</b> attribute                                                            |
| R16  | <b>MERCHANTS – HAS –</b><br><b>ORDERS</b>                                               | 1:N             | (0,N)<br>partial /<br>(1,1) <b>total</b> |                                                                                      |
| R17  | <b>CUSTOMER – PLACES –</b><br><b>ORDERS</b>                                             | 1:N             | (0,N)<br>partial /<br>(1,1) <b>total</b> |                                                                                      |
| R18  | <b>ORDERS – HAS_ITEM –</b><br><b>ORDER_ITEMS</b> (weak/<br>identifying)                 | 1:N identifying | (0,N)<br>partial /<br>(1,1) <b>total</b> |                                                                                      |
| R19  | <b>PRODUCT_VARIANTS –</b><br><b>REFERENCES –</b><br><b>ORDER_ITEMS</b>                  | 1:N             | (0,N)<br>partial /<br>(1,1) <b>total</b> |                                                                                      |
| R20  | <b>ORDERS –</b><br><b>HAS_PAYMENT –</b><br><b>PAYMENTS</b>                              | 1:N             | (0,N)<br>partial /<br>(1,1) <b>total</b> |                                                                                      |
| R21  | <b>ORDERS –</b><br><b>HAS_SHIPMENT –</b><br><b>SHIPMENTS</b>                            | 1:N             | (0,N)<br>partial /<br>(1,1) <b>total</b> | <b>shipments.order_id</b><br>NOT NULL — every<br>shipment must belong to<br>an order |
| R22  | <b>WAREHOUSES –</b><br><b>SHIPS_FROM –</b><br><b>SHIPMENTS</b>                          | 1:N             | (0,N)<br>partial /<br>(1,1) <b>total</b> |                                                                                      |
| R23  | <b>PRODUCTS –</b><br><b>REVIEWED_AS –</b><br><b>REVIEWS</b>                             | 1:N             | (0,N)<br>partial /<br>(1,1) <b>total</b> | REVIEWS is a full entity<br>(not bridge);<br><b>product_id</b> NOT<br>NULL           |
| R23b | <b>CUSTOMER –</b><br><b>WRITTEN_BY –</b><br><b>REVIEWS</b>                              | 1:N             | (0,N)<br>partial /<br>(1,1) <b>total</b> | <b>customer_user_id</b><br>NOT NULL — every<br>review has an author                  |
| R24  | <b>DISCOUNTS –</b><br><b>APPLIED_TO – ORDERS</b><br>(bridge<br><b>discount_usages</b> ) | M:N             | (0,N)<br>partial /<br>(0,N) partial      |                                                                                      |
| R25  | <b>USERS – ACTOR_OF –</b><br><b>ACTIVITY_LOG</b>                                        | 1:N             | (0,N)<br>partial /<br>(0,1) partial      | LOG side (0,1): system<br>events have null<br><b>actor_user_id</b>                   |

## 5. Business Rules & Constraints

1. **Tenant isolation** — every non-global table carries `merchant_id`; composite FKs (`merchant_id, X`) prevent cross-tenant references.
2. **Global user identity** — one `users` row per person; membership in merchants is via `merchant_staff`, shopping is via `orders` keyed on `customer_user_id`.
3. **Weak-entity identity** — `PRODUCT_VARIANTS`, `ORDER_ITEMS` share identity with their owner; cannot exist without it.
4. **Every merchant has an owner** — `MERCHANTS.owner_user_id` NOT NULL at creation; enforced via `merchant_staff` with `role='owner'`.
5. **One owner transition** — changing owner requires demoting previous owner to admin (app-level check in Phase 4).
6. **Review uniqueness** — `UNIQUE (product_id, customer_user_id)` on `reviews`.
7. **Rating range** — `CHECK (rating BETWEEN 1 AND 5)` on `reviews`.
8. **Verified-purchase flag coherence** — `is_verified_purchase` TRUE iff `order_id IS NOT NULL`.
9. **Order immutability** — once `orders.status` reaches `paid`, financial fields (`subtotal`, `tax_total`, etc.) are immutable; changes go through refund/cancel flow.
10. **Line-item snapshot** — `order_items.product_title/sku/unit_price` frozen at order placement.
11. **Inventory reservation** — before an order transitions to `paid`, `inventory.quantity_reserved` must have sufficient headroom; on cancel, reservation decremented.
12. **Discount validity** — enforced at order placement: `NOW() BETWEEN starts_at AND COALESCE(ends_at, NOW())`, `is_active = TRUE`, `used_count < max_uses`.
13. **Recursive category depth** — no enforced max, but Phase 5 report notes convention of  $\leq 4$  levels.
14. **Audit immutability** — `activity_log` is append-only (app-level); no UPDATE/DELETE after insertion.
15. **Currency consistency** — an order's `currency` must match `merchants.currency` for that tenant.
16. **Cart → order transition** — on successful checkout, cart status set to `converted`; cart rows retained for analytics (not deleted).

*End of Phase 1*